

Q1. Which of the following best describes a Logical Agent in AI?

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| a) An agent that reacts randomly to the environment | c) An agent that only follows pre-defined scripts |
| b) An agent that uses a knowledge base and logical inference to decide actions | d) An agent that learns from rewards |

Q2. In Propositional Logic, ' $P \rightarrow Q$ ' is logically equivalent to:

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| a) $P \wedge \neg Q$ | c) $\neg Q \rightarrow \neg P$ only |
| b) $\neg P \vee Q$ | d) Both b and c |

Q3. Conjunctive Normal Form (CNF) requires the formula to be expressed as:

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| a) A disjunction of conjunctions | c) A conjunction of literals only |
| b) A conjunction of disjunctions (clauses) | d) A sequence of implications |

Q4. In FOL, the sentence $\forall x: \text{Student}(x) \rightarrow \text{Passes}(x)$ means:

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| a) Some students pass | c) If something passes, it is a student |
| b) Every entity that is a student passes | d) There exists at least one student who passes |

Q5. Unification in FOL is the process of finding substitutions that:

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| a) Convert clauses to CNF | c) Prove a goal using backward chaining |
| b) Make two logical expressions syntactically identical | d) Derive facts using Modus Ponens |

Q6. In the Resolution algorithm, deriving the empty clause $\square$ indicates:

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| a) The knowledge base is consistent | c) A contradiction — the original goal is proven |
| b) The negated goal is satisfiable | d) More clauses need to be added |

Q7. Forward Chaining is best described as a _____ strategy.

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| a) Goal-driven / top-down | c) Heuristic-based depth-first |
| b) Data-driven / bottom-up | d) Probabilistic inference |

Q8. Backward Chaining begins reasoning from:

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| a) All known facts toward a conclusion | c) A random clause in the knowledge base |
| b) The goal and works backward to verify supporting facts | d) The resolution refutation tree |

Q9. Which step is NOT part of converting a sentence to CNF?

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| a) Eliminate biconditionals ($\leftrightarrow$) | c) Apply Skolemization to remove existential quantifiers |
| b) Move negations inward using De Morgan's laws | d) Distribute AND over OR |

Q10. Knowledge Engineering in FOL involves:

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| a) Training a neural network on domain-specific data | c) Writing heuristic search strategies |
| b) Defining domain concepts, relations, and axioms formally in FOL | d) Encoding reward functions for reinforcement learning |

Q11. Modus Ponens states: If $P \rightarrow Q$ is known and P is true, then:

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| a) $\neg Q$ is true | c) P is uncertain |
| b) Q must be true | d) The KB is inconsistent |

Q12. The Most General Unifier (MGU) is the substitution that:

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| a) Is the most specific possible match | c) Eliminates all variables from a clause |
| b) Makes two terms identical with minimal variable binding | d) Applies resolution to produce CNF |

Q13. Which inference method is more appropriate when a specific goal is known?

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| a) Forward Chaining | c) Backward Chaining |
| b) Truth Table method | d) A* search |

Q14. Propositional Logic differs from First Order Logic because it:

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| a) Supports probabilistic reasoning | c) Is always more expressive |
| b) Cannot represent objects, relations, or quantifiers — only truth values of atomic propositions | d) Handles uncertainty better |

Q15. Resolution Refutation proves a goal G by:

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| a) Directly deriving G from the KB using forward chaining | c) Checking G in the truth table |
| b) Adding $\neg G$ to the KB and deriving a contradiction (empty clause $\square$) | d) Applying Modus Tollens repeatedly |

Q16. Learning from observation in AI refers to:

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| a) An agent receiving numerical rewards from the environment | c) Acquiring knowledge by following explicit symbolic rules |
| b) Improving performance by examining examples or experience from the environment | d) Memorizing a fixed dataset without generalization |

Q17. Inductive learning is best described as:

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| a) Deriving specific conclusions from general rules | c) Memorizing training data exactly |
| b) Generalizing a hypothesis from specific training examples | d) Applying search algorithms to find optimal policies |

Q18. In a Decision Tree, what does each internal node represent?

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| a) A class label / output decision | c) A test on an attribute / feature |
| b) A probability distribution over outcomes | d) A training data sample |

Q19. Explanation-Based Learning (EBL) differs from inductive learning because it:

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| a) Requires large amounts of training data to generalize | c) Uses prior domain knowledge to explain and generalize a single example |
| b) Learns only from numeric data using regression | d) Applies unsupervised clustering to find hidden patterns |

Q20. In statistical learning, a Naive Bayes classifier assumes:

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| a) All features are conditionally dependent on each other given the class | c) All features are conditionally independent given the class label |
| b) The data must follow a uniform distribution | d) Labels are determined by a reward function |

Q21. Learning with complete data (no hidden variables) in a Bayesian network involves:

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| a) Using the EM algorithm to estimate missing values | c) Directly computing maximum likelihood estimates from observed data |
| b) Training a deep neural network with dropout regularization | d) Applying reinforcement signals to update weights |

Q22. The Expectation-Maximization (EM) algorithm is used when:

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| a) Training data is fully labeled and no variables are hidden | c) The dataset contains hidden (latent) variables that cannot be directly observed |
| b) An agent explores an unknown environment using trial and error | d) Backpropagation fails to converge in a neural network |

Q23. In a Neural Network, the activation function serves to:

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| a) Initialize the weights of each neuron to zero | c) Introduce non-linearity to enable learning of complex patterns |
| b) Normalize the input dataset before training | d) Store previously learned examples in memory |

Q24. Which of the following correctly describes backpropagation in a neural network?

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| a) Weights are updated in the forward pass using input gradients | c) The error is propagated backward from output to input layers to adjust weights |
| b) Outputs of each layer are stored and reused in later epochs | d) A separate model is trained for each hidden layer independently |

Q25. A Support Vector Machine (SVM) — a type of kernel machine — classifies data by:

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| a) Minimizing the number of training examples used | c) Finding the maximum-margin hyperplane that separates classes |
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b) Grouping similar data points into clusters using centroids

d) Updating weights using Q-values from an environment

Q26. The kernel trick in kernel machines (SVMs) allows:

a) Reducing the number of support vectors needed

c) Operating in a high-dimensional feature space without explicitly computing coordinates

b) Accelerating gradient descent convergence

d) Replacing hidden layers in a deep neural network

Q27. In Reinforcement Learning, the agent aims to maximize:

a) Classification accuracy on a fixed test set

c) Cumulative reward received over time through interaction with the environment

b) The number of features selected from the training data

d) The size of the neural network hidden layer

Q28. The Q-Learning algorithm in Reinforcement Learning is classified as:

a) A supervised learning algorithm requiring labeled datasets

c) A model-free, off-policy temporal difference learning method

b) An unsupervised clustering technique using centroids

d) A decision-tree-based ensemble classifier

Q29. Overfitting in machine learning occurs when:

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| a) The model performs poorly on both training and test data | c) The model fits training data too closely and generalizes poorly to new data |
| b) The learning rate is set too low during gradient descent | d) The training set has more features than training examples |

Q30. Which of the following best distinguishes Reinforcement Learning from Supervised Learning?

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| a) RL uses labeled input-output pairs; SL uses reward signals | c) RL learns via trial-and-error with reward feedback; SL learns from labeled examples |
| b) Both RL and SL require explicit reward functions | d) RL is always faster to converge than SL on any dataset |

ANSWERS:

Q	Answer	Correct Option	Quick Reason
1	b	Knowledge base + logical inference	Logical agents reason over a KB to select actions.
2	d	Both b and c	$(P \rightarrow Q) \equiv (\neg P \vee Q)$, and its contrapositive is $(\neg Q \rightarrow \neg P)$.
3	b	Conjunction of disjunctions	CNF = AND of OR clauses.
4	b	Every student passes	$(\forall x (\text{Student}(x) \rightarrow \text{Passes}(x)))$.
5	b	Make expressions syntactically identical	This is the purpose of unification.
6	c	Contradiction $\rightarrow$ goal proven	In resolution refutation, deriving $(\square)$ proves the original goal.
7	b	Data-driven / bottom-up	Forward chaining starts from known facts.
8	b	Goal and works backward	Backward chaining is goal-driven.
9	c	Skolemization	Skolemization is specifically part of FOL-to-CNF conversion, not ordinary propositional CNF conversion.
10	b	Define domain concepts, relations and axioms	This is knowledge engineering in FOL.
11	b	Q must be true	From $(P \rightarrow Q)$ and (P) , infer (Q) .

Q	Answer	Correct Option	Quick Reason
12	b	Minimal variable binding	MGU makes expressions identical with the most general substitution.
13	c	Backward Chaining	Best when a specific goal is given.
14	b	Only truth-valued propositions	Propositional logic lacks variables, relations and quantifiers.
15	b	Add $\neg G$ and derive $\square$	This is resolution refutation.
16	b	Learn from examples/experience	Learning from observation improves performance from experience.
17	b	Generalize from examples	This is inductive learning.
18	c	Test on attribute/feature	Internal decision-tree nodes perform attribute tests.
19	c	Uses prior domain knowledge	EBL explains an example using domain knowledge and generalizes it.
20	c	Features conditionally independent	This is the "naive" assumption.
21	c	Directly compute MLE	Complete observed data allows direct parameter estimation.
22	c	Hidden/latent variables	EM estimates parameters when latent variables are involved.
23	c	Introduces non-linearity	Activation functions allow neural networks to learn complex patterns.
24	c	Error propagated backward	Backpropagation computes gradients backward to update weights.
25	c	Maximum-margin hyperplane	This is the fundamental idea of SVM classification.
26	c	High-dimensional feature space	Kernel trick computes similarities without explicitly mapping coordinates.
27	c	Cumulative reward	RL seeks to maximize expected cumulative reward.
28	c	Model-free, off-policy TD	Q-learning is a model-free, off-policy temporal-difference method.
29	c	Fits training data too closely	Overfitting causes poor generalization.
30	c	Trial-and-error vs labeled examples	RL learns from reward feedback; supervised learning uses labeled data.

